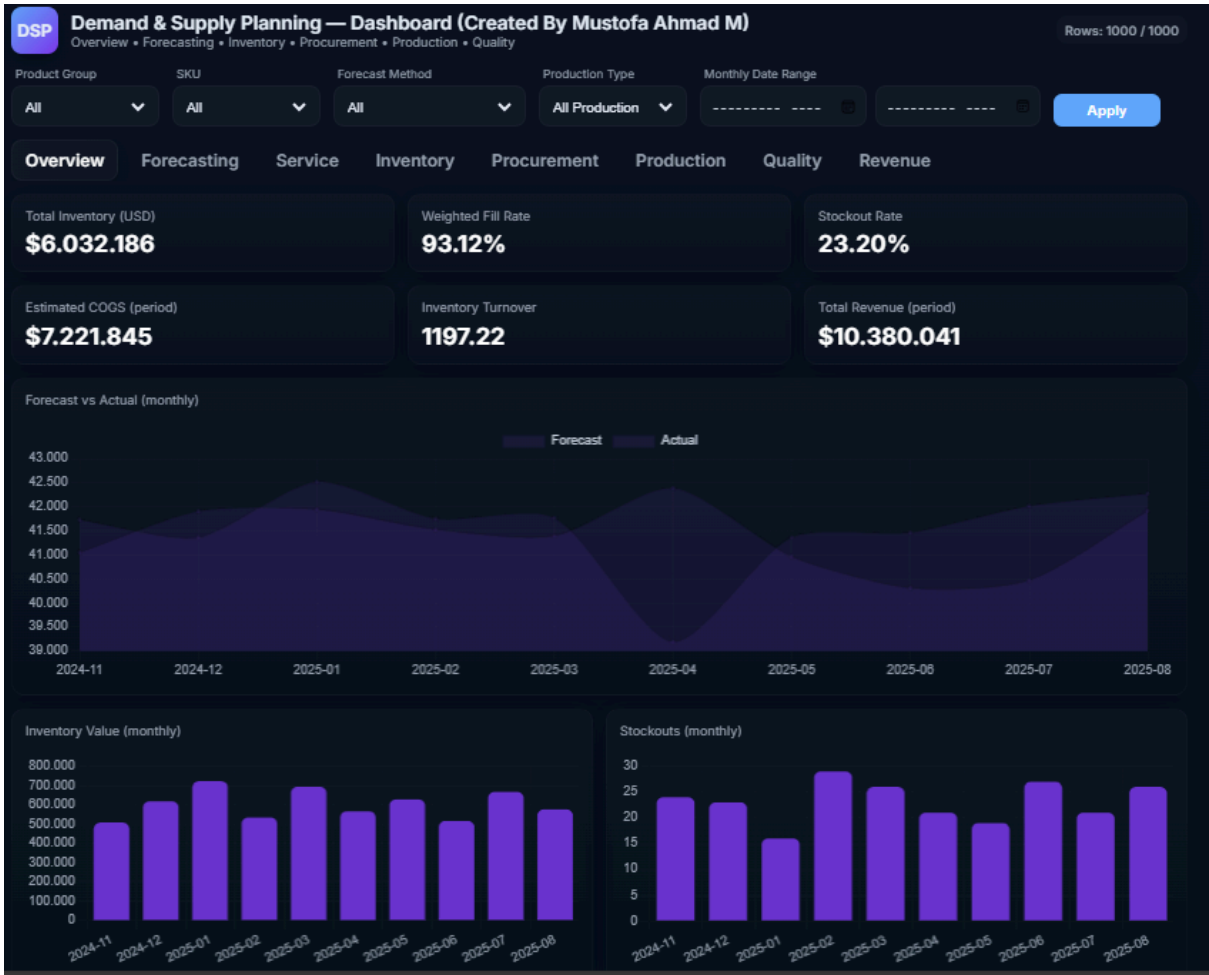


Demand & Supply Planning Dashboard

Role Play: Demand & Supply Planning

Subject: Overview and functional guide for the new Demand & Supply Planning (DSP) Dashboard, a strategic tool for managing our end-to-end integrated business planning (IBP) cycle. This document outlines the capabilities of our new Demand & Supply Planning Dashboard. This case for the industrial chemical sector, effective planning is critical for balancing service levels, managing working capital, and ensuring production efficiency. This dashboard was developed to transform our monthly S&OP/IBP cycle from a series of disconnected spreadsheets into a dynamic, integrated analytical platform. It is designed to provide clear visibility into forecast accuracy, inventory health, supply risks, and production alignment, enabling us to make more agile and profitable planning decisions.



1. Data Set Overview

- **Replenishment_Order_units**: The suggested order quantity based on planning parameters.
- **Production_Plan_units**: The planned production volume for in-house manufactured items.

Financial & Health Metrics

- **Unit_Cost_USD, Inventory_Value_USD**: The cost per unit and the total financial value of on-hand inventory.
- **Days_of_Inventory**: A measure of how long the current on-hand inventory will last based on average demand.
- **Fill_Rate_pct, Stockout_Flag**: Metrics indicating our ability to meet customer demand from available stock.
- **SLOB_Flag** (Slow-Moving and Obsolete): A flag for inventory that is at high risk of becoming obsolete.
- **Shelf_Life_days, Expiry_Date, Days_to_Expiry**: Critical data for managing products with a limited shelf life.

2. Key Metrics & Measurements

The dashboard is structured into functional modules (tabs) that align with the core pillars of the Demand & Supply Planning process.

A. Forecasting Performance

- **Weighted MAPE (Mean Absolute Percentage Error):**
 - **Objective:** To measure the overall accuracy of our demand forecast, weighted by volume to prioritize the most significant products.
 - **Goal:** To achieve a W-MAPE of **less than 15%**. A lower MAPE indicates a more accurate and reliable demand plan, which is the foundation for all subsequent supply planning decisions.
- **Forecast Bias:**
 - **Objective:** To identify systemic tendencies to either over-forecast (negative bias) or under-forecast (positive bias).
 - **Goal:** To maintain a forecast bias **close to zero**. A neutral bias indicates that our forecasting errors are random, not systemic, which allows for more reliable safety stock calculations.

B. Service & Fulfillment

- **Weighted Fill Rate:**
 - **Objective:** To measure our ability to fulfill actual customer demand from the available supply (on-hand + planned receipts) in a given month.
 - **Goal:** To maintain a Weighted Fill Rate of **over 98%** for our 'A' class items. This KPI is a direct measure of customer service and our ability to prevent lost sales due to stockouts.
- **Stockout Rate:**

- **Objective:** To track the percentage of SKU-months where a stockout occurred (i.e., available supply was insufficient to cover actual demand).
- **Goal:** To minimize the stockout rate to **less than 2%**. This metric highlights critical failures in the supply plan that lead to direct revenue loss.

C. Inventory Management

- **Days of Inventory (Value-Weighted):**
 - **Objective:** To measure the average number of days of inventory we hold, weighted by value to accurately reflect our working capital investment.
 - **Goal:** To optimize our inventory levels, balancing service with cost, and achieve a target **Days of Inventory of under 45 days**.
- **SLOB Value (Slow-Moving & Obsolete):**
 - **Objective:** To quantify the financial value of inventory that is at high risk of becoming obsolete.
 - **Goal:** To proactively manage and reduce the SLOB value through targeted sales efforts or responsible disposal, keeping it **below 1% of total inventory value**.

3. Key Performance Indicators (KPIs)

As the Demand & Supply Planning Manager, the primary responsibility is to balance the trade-offs between customer service, inventory costs, and operational efficiency. The following KPIs are our most critical measures of success.

- **KPI 1: Weighted MAPE (Mean Absolute Percentage Error)**
 - **Definition:** The volume-weighted average of forecast errors, expressed as a percentage.
 - **Importance:** This is the cornerstone KPI for the demand planning function. Its accuracy dictates the quality of every decision made downstream in procurement, production, and inventory management. A low W-MAPE leads to optimized inventory and higher service levels.
 - **Strategic Goal:** **Achieve and maintain a W-MAPE of < 15% for the rolling 3-month period.**
- **KPI 2: Weighted Fill Rate**
 - **Definition:** The percentage of actual customer demand (in units) that was successfully fulfilled from available supply.
 - **Importance:** This is the ultimate measure of our supply plan's effectiveness from the customer's perspective. It directly translates our planning efforts into service level reality and revenue capture.
 - **Strategic Goal:** **Maintain a Weighted Fill Rate of $\geq 98\%$ for strategic product groups.**
- **KPI 3: Days of Inventory (Value-Weighted)**
 - **Definition:** The average number of days that inventory is held before being sold, weighted by its financial value.
 - **Importance:** This KPI is a primary measure of working capital efficiency. Optimizing DOI frees up cash, reduces holding costs, and minimizes the risk of obsolescence, directly impacting the company's financial health.

- **Strategic Goal:** Reduce the value-weighted Days of Inventory by 10% year-over-year while maintaining service level targets.
- **KPI 4: Revenue at Risk (due to Stockouts)**
 - **Definition:** The total potential revenue lost in a given period due to insufficient stock to meet actual customer demand.
 - **Importance:** This KPI translates supply planning gaps into a direct, tangible financial impact. It provides a powerful justification for investments in improving forecast accuracy and optimizing safety stock levels.
 - **Strategic Goal:** Keep monthly Revenue at Risk to less than 1% of total monthly revenue.

4. Dashboard Functionality Guide [\(Click Here To Preview Dashboard\)](#)

The Demand & Supply Planning Dashboard is an integrated tool designed to provide a comprehensive view of the entire planning cycle.

A. Master Controls & Filters

- **Filter Bar:** Located at the top, this panel allows you to dynamically segment all dashboard data by **Product Group**, **SKU**, **Forecast Method**, **Production Type (In-House vs. Purchased)**, and a **Monthly Date Range**.
- **Apply Button:** Executes the data refresh across all dashboard components based on your selected criteria.

B. Navigation Tabs The dashboard's core functionality is organized into logical tabs that mirror the S&OP/IBP process.

- **Overview:** The executive summary view, showing high-level KPIs like Total Inventory Value, Weighted Fill Rate, and Stockout Rate, alongside key trend charts for forecast vs. actuals.
- **Forecasting:** A deep dive into demand planning performance, with KPIs like Weighted MAPE and Forecast Bias. Charts visualize the monthly MAPE trend and the composition of forecast error by product group.
- **Service:** Focuses on customer-facing metrics. It tracks the Weighted Fill Rate, Stockout Rate, and PO On-Time performance from suppliers.
- **Inventory:** The central hub for monitoring working capital and inventory health. It displays KPIs like Days of Inventory, SLOB value, and includes charts for safety stock scenario comparisons and inventory value composition.
- **Procurement:** Analyzes the supply side of the plan, with metrics on Average Supplier Lead Time and the percentage of times on-hand inventory has breached the reorder point.
- **Production:** Provides a view of in-house manufacturing plans, tracking the total planned production volume and the average capacity utilization.
- **Quality:** Focuses on inventory longevity and risk, displaying metrics and charts related to the age of inventory and days to expiry.

- **Revenue:** Connects the supply plan to financial outcomes, tracking total revenue and, crucially, the "Revenue at Risk" due to potential stockouts.

C. KPI Cards & Charts Each tab is populated with two primary visualization components:

- **KPI Grid:** A set of "cards" at the top of each page displays the most critical metrics for that specific planning function, providing an immediate, at-a-glance summary of performance.
- **Charts:** Each tab features one or more charts designed to provide deeper visual insights. For example:
 - The **Overview** tab features a line chart comparing **Forecast vs. Actual Demand** over time, which is the foundational chart for any planning review.
 - The **Inventory** tab includes a doughnut chart showing the **Inventory Value by Product Group**, helping to visualize where working capital is concentrated.
 - The **Forecasting** tab includes a line chart that tracks the **Monthly MAPE Trend**, allowing you to monitor the continuous improvement of forecast accuracy.